STUDENT COPY — NO SOLVED ANSWERS

Vyom · Is hafte ka print pack

7 Dec – 13 Dec 2026 · Monday–Sunday · NDPS Class 9

Naam: Vyom | Week tasks: 20 | Scheduled tests: 1

Sunday print handover: Sunday, 6 December 2026. Agle Monday se is pack ke daily pages follow karo. Roz tasks tick karo; tests par naam/date aur answers likho. Answer key parent ke paas rahegi.

Din / date	Kaam	Tests / focus
Monday, 7 December 2026	2 tasks	Maths Pre-annual mock (80 marks)
Tuesday, 8 December 2026	3 tasks	Pre-annual exams (7–20 Dec)
Wednesday, 9 December 2026	3 tasks	Pre-annual exams (7–20 Dec)
Thursday, 10 December 2026	3 tasks	Pre-annual exams (7–20 Dec)
Friday, 11 December 2026	3 tasks	Pre-annual exams (7–20 Dec)
Saturday, 12 December 2026	3 tasks	Pre-annual exams (7–20 Dec)
Sunday, 13 December 2026	3 tasks	Pre-annual exams (7–20 Dec)

Schedule v4 · 2026-10-12 · Sanskrit coverage. Daily blocks original shifted routine ke mutabik hain; asli weekday date har page par di hai.

School ki subject-wise exam date sheet/paper pattern abhi confirm karna baaki hai. Exam-period sample-section tasks ko official school papers mat samjhein.

Monday, 7 December 2026

Pre-annual revision week

	Samay	Subject	Aaj kya karna hai
☐	9:00–12:00	**Maths**	Test Maths pre-annual mock paper (3 hr, 80 marks) — exam jaisa timed; checking kal tuition me
☐	Shaam (45 min)	**Maths**	Correction Mock ki galtiyan Mistakes notebook me + date sheet ke pehle paper ki revision

Ho gaye: ___ / 2 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Maths · Pre-Annual Mock Paper · 7 Dec

Class 9 · Ganita Manjari Ch 1–13 (Ch 14 Surface Area & Volume chhod kar)

Test ID: T2026-12-07 · Monday, 7 December 2026

Naam: _____ Samay: 180 min Marks: ___ / 80

Slot: 9:00–12:00

Paper mein 38 prashn aur 5 section hain (A–E). Saare prashn compulsory hain; Section D mein 2 prashnon mein internal choice hai.
Section A: 1 mark · B: 2 marks · C: 3 marks · D: 5 marks · E: case-based 4 marks.
Jahan zaroori ho π = 22/7 lo. Har step likho — steps ke marks hain.
Note (parent): school ka pre-annual syllabus aate hi match kar lein — jo chapter syllabus mein na ho, uske prashn chhod dein aur marks us hisaab se ghata dein.

Section A · MCQ + Assertion-Reason · 1 mark each

1. The point (–3, 5) lies in the: **[1]**

(a) I quadrant (b) II quadrant
(c) III quadrant (d) IV quadrant

Answer space · T2026-12-07 · Q1

2. The distance between the points (2, –4) and (2, 7) is: **[1]**

(a) 3 (b) 9
(c) 11 (d) 13

Answer space · T2026-12-07 · Q2

3. The degree of the polynomial $3y^3 - 4y^2 + 5$ is: **[1]**

(a) 1 (b) 2
(c) 3 (d) 5

Answer space · T2026-12-07 · Q3

4. The slope of the line y = –2x + 7 is: **[1]**

(a) 7 (b) –2
(c) 2 (d) –7

Answer space · T2026-12-07 · Q4

5. Which of the following is irrational? **[1]**

(a) $\sqrt{16}$ (b) 0.121212...

(c) $\sqrt{7}$ (d) 22/7

Answer space · T2026-12-07 · Q5

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6. Using a suitable identity, 101^2 = **[1]**

(a) 10101 (b) 10201

(c) 10001 (d) 10200

Answer space · T2026-12-07 · Q6

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7. If $x + 1/x = 5$, then $x^2 + 1/x^2$ = **[1]**

(a) 25 (b) 27

(c) 23 (d) 10

Answer space · T2026-12-07 · Q7

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8. A chord of length 16 cm is drawn in a circle of radius 10 cm. Its distance from the centre is: **[1]**

(a) 6 cm (b) 8 cm

(c) 4 cm (d) 12 cm

Answer space · T2026-12-07 · Q8

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9. The angle subtended by a diameter at any point on the circle is: **[1]**

(a) 45° (b) 60°

(c) 90° (d) 180°

Answer space · T2026-12-07 · Q9

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10. The area of a sector of radius 7 cm and central angle 90° is: **[1]**

(a) 38.5 cm^2 (b) 154 cm^2

(c) 11 cm^2 (d) 77 cm^2

Answer space · T2026-12-07 · Q10

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11. A die is thrown once. The probability of getting a number greater than 4 is: **[1]**

(a) 1/6 (b) 1/3

(c) 1/2 (d) 2/3

Answer space · T2026-12-07 · Q11

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12. The common ratio of the GP 5, –10, 20, –40, ... is: **[1]**

(a) 2 (b) –2

(c) –5 (d) 1/2

Answer space · T2026-12-07 · Q12

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13. The 10th term of the AP 2, 7, 12, 17, ... is: **[1]**

(a) 45 (b) 47

(c) 52 (d) 50

Answer space · T2026-12-07 · Q13

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14. The converse of "If a triangle is equilateral, then it is isosceles" is: **[1]**

(a) If a triangle is not equilateral, then it is not isosceles.

(b) If a triangle is isosceles, then it is equilateral.

(c) If a triangle is isosceles, then it is not equilateral.

(d) A triangle is equilateral only if it is scalene.

Answer space · T2026-12-07 · Q14

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15. Riya scored 80 in a test of weight 3 and 60 in a test of weight 1. Her weighted average is: **[1]**

(a) 70 (b) 75

(c) 72 (d) 65

Answer space · T2026-12-07 · Q15

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16. The line $x + 2y = 6$ cuts the y-axis at: **[1]**

(a) (6, 0) (b) (0, 6)

(c) (0, 3) (d) (3, 0)

Answer space · T2026-12-07 · Q16

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17. An algorithm checks every number j from 1 to n to see whether j divides n. For n = 20, how many numbers does it check? **[1]**

(a) 10

(b) 20

(c) 6

(d) 4

Answer space · T2026-12-07 · Q17

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18. In a parallelogram ABCD, $\angle A = 70^\circ$. Then $\angle C$ = **[1]**

(a) 110° (b) 70°

(c) 20° (d) 140°

Answer space · T2026-12-07 · Q18

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19. **A:** $(x + y)^2 = x^2 + y^2$ is an identity. **[1]**

R: An identity is an equation that is true for all values of the variables.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Answer space · T2026-12-07 · Q19

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20. **A:** The centroid of a triangle divides each median in the ratio 2 : 1. **[1]**

R: The three medians of a triangle are concurrent.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Answer space · T2026-12-07 · Q20

Section B · Very Short Answer · 2 marks each

21. Points A(–3, 4) and B(5, 4) are given. Find the length AB and the point on AB that is equidistant from A and B. **[2]**

Answer space · T2026-12-07 · Q21

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22. (i) Factorise $4x^2 – 12x + 9$. (ii) Evaluate 98×102 using a suitable identity. **[2]**

Answer space · T2026-12-07 · Q22

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23. Find two rational numbers between 1/3 and 1/2. **[2]**

Answer space · T2026-12-07 · Q23

24. A bag has 5 red, 3 green and 2 yellow balls. One ball is drawn at random. Find (i) P(green) (ii) P(not yellow). **[2]**

Answer space · T2026-12-07 · Q24

25. Write the converse of: "If two angles are vertically opposite, then they are equal." Is the converse true? Justify. **[2]**

Answer space · T2026-12-07 · Q25

Section C · Short Answer · 3 marks each

26. Prove that $\sqrt{2}$ is irrational. **[3]**

Answer space · T2026-12-07 · Q26

27. Find the area of a triangle whose sides are 13 cm, 14 cm and 15 cm using Heron's formula. **[3]**

Answer space · T2026-12-07 · Q27

28. Rahul saves ₹50 in week 1, ₹65 in week 2, ₹80 in week 3, and so on. (i) How much does he save in week 20? (ii) In which week will he first save more than ₹500? **[3]**

Answer space · T2026-12-07 · Q28

29. AB and CD are parallel chords of a circle of radius 13 cm, on opposite sides of the centre. AB = 10 cm and CD = 24 cm. Find the distance between the chords. **[3]**

Answer space · T2026-12-07 · Q29

30. Class A has 30 students with average marks 72, and Class B has 20 students with average 82. (i) Find the combined average. (ii) A new student scoring 95 joins Class B. Find the new average of Class B. **[3]**

Answer space · T2026-12-07 · Q30

31. Solve by the method of elimination: $2x + 3y = 13$ and $3x - 2y = 0$. **[3]**

Answer space · T2026-12-07 · Q31

Section D · Long Answer · 5 marks each

32. Draw the graphs of $y = 2x - 1$ and $x + y = 5$ on the same axes. (i) Find the point where they intersect. (ii) Find the points where each line cuts the y-axis. (iii) Find the area of the triangle formed by the two lines and the y-axis. **[5]**

ATHVA / OR

The cost of 2 pens and 3 notebooks is ₹130, and the cost of 3 pens and 2 notebooks is ₹120. Form the pair of linear equations and solve them by substitution to find the cost of one pen and one notebook.

Answer space · T2026-12-07 · Q32

33. (i) Factorise $x^3 - 27$. (ii) Without calculating the cubes, find $(-12)^3 + 7^3 + 5^3$. (iii) Expand $(2x + 3y)^3$. **[5]**

Answer space · T2026-12-07 · Q33

34. Prove that the quadrilateral formed by joining the midpoints of the sides of any quadrilateral (in order) is a parallelogram. **[5]**

ATHVA / OR

Prove that the diagonals of a parallelogram bisect each other.

Answer space · T2026-12-07 · Q34

35. A spinner has three equal parts coloured Red, Blue and Green. It is spun twice. (i) Draw a tree diagram and write the sample space. Find the probability of getting (ii) the same colour both times (iii) at least one Red (iv) no Green. **[5]**

Answer space · T2026-12-07 · Q35

Section E · Case-Based · 4 marks each

36. **[4]**

> A park is drawn on a grid with the main gate at the origin O(0, 0). The fountain is at F(4, 3), a bench at B(–2, 3) and a lamp post at L(4, –5). 1 unit = 10 m.

(i) In which quadrant is the lamp post? [1]
(ii) Find the distance FB. [1]
(iii) Find the area of triangle FBL in square units and in m^2. [2]

Answer space · T2026-12-07 · Q36

37. **[4]**

> A school gives final marks as a weighted average: Unit test 20%, Half-yearly 30%, Annual 50%. Riya scored 80 in the unit test, 70 in the half-yearly and 90 in the annual exam.

(i) Find Riya's final (weighted) score. [1]
(ii) Find the simple average of her three scores. [1]
(iii) Aman scored 80 and 70 in the first two exams. What must he score in the annual exam to get a final score of 85? [2]

Answer space · T2026-12-07 · Q37

38. **[4]**

> In a Sierpiński triangle, we start (stage 0) with one shaded triangle. At every stage, each shaded triangle is replaced by 3 smaller shaded triangles, and the shaded area becomes ¾ of the shaded area of the previous stage.

(i) How many shaded triangles are there at stage 3? [1]
(ii) Write the number of shaded triangles at stage n. What kind of progression is this? [1]
(iii) If the shaded area at stage 0 is 64 cm^2, find the shaded area at stage 3. [2]

Answer space · T2026-12-07 · Q38

Teacher/parent: marks answer key ke mutabik; galti C concept / R recall / K careless / T time.

Tuesday, 8 December 2026

Pre-annual exams (7–20 Dec) · Pre-annual exam

	Samay	Subject	Aaj kya karna hai
☐	Paper ke baad	**Sab**	Correction Aaj ke paper me jo galat hua — 10 min me note karo
☐	Dopahar + shaam	**Sab**	Revision Agle paper ki revision (date sheet dekh kar): important Q, diagrams, formulas, maps
☐	Raat	**Sab**	Recall 15 min quick recall; 10 baje tak so jao

Ho gaye: ___ / 3 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Wednesday, 9 December 2026

Pre-annual exams (7–20 Dec) · Pre-annual exam

	Samay	Subject	Aaj kya karna hai
☐	Paper ke baad	**Sab**	Correction Aaj ke paper me jo galat hua — 10 min me note karo
☐	Dopahar + shaam	**Sab**	Revision Agle paper ki revision (date sheet dekh kar): important Q, diagrams, formulas, maps
☐	Raat	**Sab**	Recall 15 min quick recall; 10 baje tak so jao

Ho gaye: ___ / 3 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Thursday, 10 December 2026

Pre-annual exams (7–20 Dec) · Pre-annual exam

	Samay	Subject	Aaj kya karna hai
☐	Paper ke baad	**Sab**	Correction Aaj ke paper me jo galat hua — 10 min me note karo
☐	Dopahar + shaam	**Sab**	Revision Agle paper ki revision (date sheet dekh kar): important Q, diagrams, formulas, maps
☐	Raat	**Sab**	Recall 15 min quick recall; 10 baje tak so jao

Ho gaye: ___ / 3 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Friday, 11 December 2026

Pre-annual exams (7–20 Dec) · Pre-annual exam

	Samay	Subject	Aaj kya karna hai
☐	Paper ke baad	**Sab**	Correction Aaj ke paper me jo galat hua — 10 min me note karo
☐	Dopahar + shaam	**Sab**	Revision Agle paper ki revision (date sheet dekh kar): important Q, diagrams, formulas, maps
☐	Raat	**Sab**	Recall 15 min quick recall; 10 baje tak so jao

Ho gaye: ___ / 3 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Saturday, 12 December 2026

Pre-annual exams (7–20 Dec) · Pre-annual exam

	Samay	Subject	Aaj kya karna hai
☐	Paper ke baad	**Sab**	Correction Aaj ke paper me jo galat hua — 10 min me note karo
☐	Dopahar + shaam	**Sab**	Revision Agle paper ki revision (date sheet dekh kar): important Q, diagrams, formulas, maps
☐	Raat	**Sab**	Recall 15 min quick recall; 10 baje tak so jao

Ho gaye: ___ / 3 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
7:00–9:00 PM	Swimming
9:00–9:30 PM	Wapas aana + fresh hona + dinner
9:30–9:45 PM	School bag, uniform aur agle din ki tayyari
9:45–10:00 PM	Relax; screen band, sone ki tayyari. Naya study block nahi.
10:00 PM	Sleep; 6:30 AM uthna (8.5 ghante). Pending kaam ke liye neend mat kaato.

Ye routine study-task ticks se alag hai. Agar travel/dinner ka actual time zyada ho, parent routine adjust karein; swimming aur sleep ke beech extra study nahi.

Aaj kya mushkil laga / chhoota kaam:

Sunday, 13 December 2026

Pre-annual exams (7–20 Dec) · Pre-annual exam

	Samay	Subject	Aaj kya karna hai
☐	Paper ke baad	**Sab**	Correction Aaj ke paper me jo galat hua — 10 min me note karo
☐	Dopahar + shaam	**Sab**	Revision Agle paper ki revision (date sheet dekh kar): important Q, diagrams, formulas, maps
☐	Raat	**Sab**	Recall 15 min quick recall; 10 baje tak so jao

Ho gaye: ___ / 3 Homework: ___ min Parent sign: _____

6:20 PM ke baad · evening routine

6:20–7:00 PM	Snack + aaram; swimming ki tayyari aur travel
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Aaj kya mushkil laga / chhoota kaam:

Extra answer sheet · 7 Dec – 13 Dec

Naam: _____ Date: _____ Subject: _____

Test ID / question number: _____

Diagram / graph aur Sunday review

Test ID / question number: _____



Is hafte ki 3 galtiyan aur sudhaar:

Week tasks: ___ / 20

Parent sign: _____

Agla hafta: sabse zaroori weak topic / pending kaam: